The Harada–Norton extension: an explicit incidence certificate

Human Compression project

11 October 2026

Abstract

We give a deterministic permutation certificate for a regular polytope of type $\{12, 5\}$ and a regular rank-three hypertope with triangular diagram $(12, 12, 5)$, both with automorphism group $HN : 2$. This supplies the Harada–Norton case omitted from the case discussion accompanying Theorem 5.3 of Piedade–Tranchida. The named representation and the complete maximal-subgroup classification remain explicit external inputs.

1 Scope and group identification

Throughout, $G = \mathrm{Aut}(HN) = HN : 2$, in the Atlas permutation representation of degree 1 140 000. We retain the identification of that representation, the simplicity of its socle HN, the outer order two, and the completeness of the maximal-subgroup list below. These are classical Atlas inputs. Checking some standard-generator identities does not reprove the identification or the classification.

Let c, d be the hash-bound Atlas standard generators, with c in the outer class $2C$ and d in class $5A$. All six tests printed on the Atlas page are checked exactly:

$$|c| = 2, \quad |d| = 5, \quad |cd| = 42, \quad |z| = 60, \quad |z^{30}d| = 22, \quad |z^{30}d^2(z^{30}d)^3| = 22, \qquad z = cd^3(cd)^4.$$

The inner involution used below is the printed standard-generator adapter

$$a_0 = (cd)^{-3}(cdcdcdcddcdcddcdd)^{10}(cd)^3.$$

The corresponding inner generator $b_0 = (cd^2)^8(cdcd^2)^5(cd^2)^{10}$ satisfies the additional checked orders

$$|a_0| = 2, \quad |b_0| = 3, \quad |a_0b_0| = 22, \quad |a_0b_0a_0b_0^2| = 5.$$

No full order enumeration of this million-point group is required by the certificate.

The source is arXiv:2605.04646v2, dated 28 May 2026. Its literal Theorem 5.3 includes HN, although the subsequent proof discusses only nine other socles. The present construction completes this additional case. It does not resolve either of the stated Fischer exceptions, and it carries no claim of priority for the existence assertion. The previously frozen nine-case packet, including the corrected O'Nan proof route, remains separate and unchanged.

2 The marking and its generation

Words are evaluated from left to right, and $x^h = h^{-1}xh$. In `HN_compact.g`, signed letters $1, 2$ denote c, d and negative letters their inverses. The file gives one word w_a and eight words w_j. Put

$$t = c, \qquad a = a_0^{w_a(c,d)}, \qquad h_j = w_j(c, d), \qquad x_j = tt^{h_j}, \qquad n_j = |x_j|.$$

All eight n_j are odd. Define

$$z_j = h_j x_j^{(n_j-1)/2}, \qquad b = (z_5 z_6 z_3 z_6 z_7 z_6 z_5 z_2 z_3 z_7)^6.$$

The usual odd-commutator identity gives $[t, z_j] = 1$: if $x = tt^h$ and $n = 2k+1 = |x|$, then $txt = x^{-1}$ and

$$t^{hx^k} = x^{-k}(tx)x^k = tx^{2k+1} = t.$$

This is Bray's centralizer construction used only to specify elements, not a claim that these eight words generate the full centralizer. The checker also verifies each commutation directly. It verifies that t, a, b are distinct nonidentity involutions and that their pair-product orders, in positions $(01, 02, 12)$, are $(12, 2, 5)$.

Lemma 1. *A subgroup of G containing c and elements of orders* 11 *and* 19 *equals G.*

Proof. The complete retained list of maximal subgroups of G is as follows. The notation describes group extensions and does not assert splitting where none is indicated.

Maximal subgroup	Order
HN	273030912000000
S_{12}	479001600
$4.HS.2$	354816000
$U_3(8) : 6$	33094656
$2_+^{1+8}.(A_5 \times A_5).2^2$	7372800
$5 : 4 \times U_3(5) : 2$	5040000
$5^{1+4}.2^{1+4}.5.4.2$	4000000
$2^6.U_4(2).2$	3317760
$(S_6 \times S_6).2^2$	2073600
$2^{3+2+6}.(3 \times L_3(2)).2$	2064384
$5^{2+1+2}.4.A_5.2$	1500000
$3^4 : 2.(S_4 \times S_4).2$	186624
$3^{1+4} : 4.S_5$	116640

Only the first order is divisible by $11 \cdot 19$. A proper subgroup containing elements of both orders would therefore lie in HN. But $c \notin HN$. This is impossible. □

The checker records all thirteen corresponding CTblLib identifiers and orders, verifies the divisibility assertion, and tests two explicit saved words in (t, a, b) for exact orders 11 and 19. Their lengths before elementary word cancellation are respectively 100 and 69. Lemma 1 therefore proves generation. The classification is the logical input; agreement of the table entries with the printed orders is a reproducibility check, not a new proof that the list is complete.

Put $s = (t, a, a^b)$. The checker verifies its pair orders $(12, 12, 5)$. Moreover

$$b = \big(a(a^b)\big)^2 a,$$

since $|ab| = 5$. Thus s generates exactly the same group. This identity establishes generation only; the incidence conditions are checked separately.

3 Exact local tests and the geometry

Let r denote either of the two triples, and write $G_I = \langle r_i : i \in I \rangle$, $I \subseteq \{0,1,2\}$. The checker constructs every proper standard subgroup and verifies

$$G_I \cap G_J = G_{I \cap J} \tag{1}$$

for all $7^2 = 49$ ordered pairs of proper subsets. The fifteen remaining pairs involve the full set and are identities because $G_I \leq G$. Thus the test establishes the full intersection property, with no assumption replacing the unchecked pairs.

Put $P_i = G_{\{0,1,2\}\setminus\{i\}}$. For each i, taking the other indices in increasing order, we check

$$(P_j P_k) \cap P_i = (P_j \cap P_i)(P_k \cap P_i). \tag{2}$$

The right side is automatically contained in the left. To test the converse, the program enumerates every ordered pair $(u, v) \in P_j \times P_k$ and checks that $uv \in P_i$ implies membership in the right side. This tests actual product sets, not subgroups generated by them. There are 376 pair tests for the string marking and 1056 for the triangular marking, in addition to their respective intersection tests. The very large ambient group never needs to be enumerated for these local conditions.

For clarity, the flag criterion has the following direct proof. Use left cosets for the incidence system. Translate an incident triple of cosets so its i-element is P_i. The other two elements have representatives $x, y \in P_i$, and their incidence says $x^{-1}y \in P_j P_k$. Equation (2) writes $x^{-1}y = uv$ with $u \in P_j \cap P_i$, $v \in P_k \cap P_i$. Then $xu = yv^{-1}$ is a common representative of all three cosets. Hence every flag has a common representative and the system is flag-transitive.

The intersection property identifies the stabilizers of all standard residues. Each panel has two chambers because its remaining singleton subgroup has order two. Each rank-two residue is connected: its two marked involutions generate its dihedral chamber group. The whole geometry is connected because the three involutions generate G. Thus the coset system is thin and residually connected. The intersection of all P_i is trivial, so the chamber action is regular and faithful. Finally, any type-preserving automorphism fixing a chamber fixes its uniquely determined adjacent chambers and, by chamber connectedness, every chamber and element. The displayed regular group is therefore the full type-preserving automorphism group.

Theorem 1. *The two saved markings define a regular polytope of type* $\{12, 5\}$ *and a regular rank-three hypertope with triangular diagram* $(12, 12, 5)$*, both with full type-preserving automorphism group* $HN : 2$*.*

Proof. Generation was proved above. The deterministic finite checks establish (1) and (2) for both markings. The first has commuting endpoints, so the retained string-C-group correspondence identifies its geometry with the regular polytope of the stated type. Every pair-product order in the second exceeds two, so its diagram is nonlinear. The preceding residue and automorphism arguments apply to both. □

4 Reproducibility and costs

The two external binary permutation files have 9 120 012 bytes each. The decoder verifies their exact SHA256 hashes, the little-endian header $(-1, 1140000, 1)$, and all 1 140 000 unsigned 64-bit images, including bijectivity. The decoded GAP inputs are generated only in the fresh output directory. The external data are not distributed with this packet; their URLs, hashes and full byte

sizes are recorded. The maximal-subgroup classification and named-group identification remain retained inputs even when all computations pass.

The complete fixed-witness checker, the compact words, the decoder and driver form the replay boundary. Search traces are construction history, not premises. Negative controls exercise decoding, the generation interface, distinctness, the intersection property, flag transitivity and the halving identity. There are no Lean declarations in this packet and no claim that GAP's permutation algorithms have been checked by a proof kernel. The manuscript, full programs and external-data requirements must all be charged when this case is incorporated into the N006 application; the word-file length alone is not the proof cost.

References

[1] C. A. Piedade and P. Tranchida, *The geometry of wreath and semi-direct products*, arXiv:2605.04646v2, 28 May 2026, Theorem 5.3.

[2] J. H. Conway, R. T. Curtis, S. P. Norton, R. A. Parker and R. A. Wilson, *Atlas of Finite Groups*, Oxford University Press, 1985.

[3] R. A. Wilson and collaborators, *Atlas of Finite Group Representations*, Harada–Norton page, `https://brauer.maths.qmul.ac.uk/Atlas/spor/HN/`.

[4] J. N. Bray, An improved method for generating the centralizer of an involution, *Arch. Math.* **74** (2000), 241–245.

[5] The GAP Group, *GAP—Groups, Algorithms, and Programming*, version 4.15.1; CTblLib 1.3.11. The exact retained files are bound in the accompanying ledger.